

new avenues of employment and digitise the economy. 5G will also open opportunities for India to connect with global markets and take advantage of economies of scale. The 5G high-level forum is drawing up an action plan in this direction. 5G once implemented, will open up new avenues of revenue for service providers.

To justify the statement, Yash adds, "Covid-19 has definitely impacted the progress. There are now signs of coming out of it amidst fresh PE investments/FDI in the sector and regulatory support, but CAPEX spend as per IDC is expected to remain muted for Indian operators over 2019 to 2024. This is largely due to the fact that any increase in material change will not immediately result in the CAPEX increase. Plus, Reliance Jio has already shown its interest in vendor-agnostic procurement, so the dynamics of the industry could shift from CAPEX-led investments to OpEx-led ones with the move towards open source and cloud-based workloads. So, once the auction takes place next year in an optimistic scenario, 5G test licenses will be followed by technology trials, industry formulation, global and regional cooperation with network sharing. So, the real commercialisation is only expected by 2022 or 2023 realistically."

According to a report of COAI, the telecom industry, along with DoT, is leaving no stone unturned in making India 5G ready, and by next year, India will be on a 5G map. To ensure the commercial deployment of 5G, a high-level 5G India forum has been instituted where experts across the globe are collaborating and discussing challenges and opportunities, making India a true 5G powered nation. Prime Minister Narendra Modi's ambitious Digital India programme and better telecom and Internet connectivity will benefit each other. 5G will open up new opportunities and catapult India's transition towards digital empowerment.

5G market forecast

The 5G economy study stated that by 2035, the entire world would understand the huge potential and benefits of 5G as it is expected to have a positive impact on various indus-

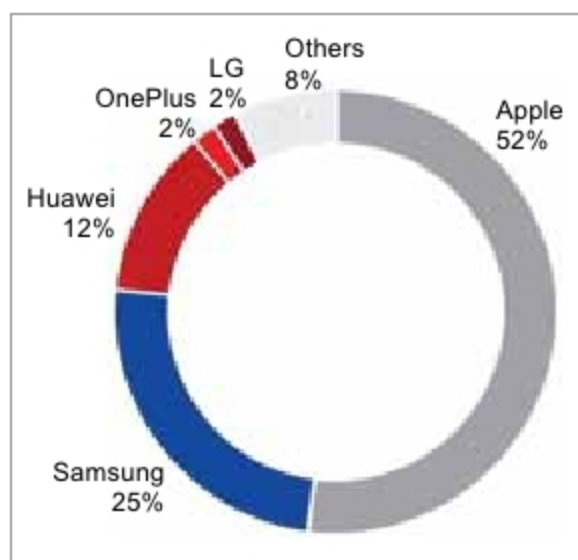


Fig. 3: Premium segment market share by OEMs, Q3 2029 (Credit: Counterpoint Technology Market Research)

trial sectors. It is likely to produce up to 12 trillion dollars worth of goods and services. In India alone, the 5G is expected to push the economy to 1 trillion dollars in the coming years, at the same time, it is now being witnessed as a potential game-changer with an ability to offer an economic growth of more than one trillion dollars by 2035. The success of 5G requires huge global investments, which is speculated to surpass 3.5 trillion dollars between 2020 and 2035.

The overall expenditure on 5G is expected to reach 88 billion dollars by 2023, according to Heavy Reading's Mobile Operator 5G Capex. The statistics above display the expected global yearly growth of investment in the telecommunication industry from 2019 to 2023. IDC expects the total 5G and LTE router/gateway market to grow by 3 billion dollars in 2024 worldwide, at a compound annual growth rate (CAGR) of 21.2 per cent. IDC projects the number of 5G connections to grow by 1.01 billion in 2023.

Yash further adds, "The domestic 5G mobile phone market is still expected to be below 50 million dollars mark in 2020. There is still a long way to go to have a mature ecosystem for justifying the CAPEX investment by operators. Jio, in its current shape and form, with all PE investments, is best poised to take a lead in the 5G space—but it could only happen with the best mix of B2B and B2C/B2B2C models. As for any technology generation to succeed, there needs to be a timely push and a timely pull from the regulatory department in the right direction, and that's what India needs."

Mathews adds, "However, initially, 5G applications and software will require a considerable amount of funding by the government. The 5GHLF has recommended an all-encompassing planning estimate of 39.34 million dollars in the first year, 52.45 million dollars in the second, 65.57 million dollars in the third, and 52.45 million dollars in the fourth year. A few roadblocks and challenges need to be addressed for 5G to become a resounding reality."

Challenges of 5G communication

The biggest challenge is of course handset cost and localised use cases. Currently, 5G smartphones are available in the price range of ₹ 40,000 and above. Due to the Covid-19 crisis, manufacturing is impacted, and so is the supply chain. Another challenge is to maintain the infrastructure in such difficult times of lockdown. Field forces faced a lot of issues in keeping towers and cables up and running. There was also a quick requirement of expanding the network capacities, which meant working round and against the clock.

According to COAI, while talking about 5G in India, the issues about the spectrum cannot be ignored. Traditionally, the spectrum has always been expensive. In India, 20MHz blocks in the 3,300-3,600MHz band for 5G services have been priced at 49.20 million dollars per MHz by TRAI in its recommendations. In Korea, for example, the same band was priced at 17.2 million dollars per MHz in auctions. The most critical factor in the uptake of the 5G spectrum would be its price. The 5G high-level forum has lately recommended early allocation of 5G spectrum, increasing the quantum of spectrum available and reducing the prices. The initial expenses on equipment would also be costly.

For 5G effective deployment, operators will have to wait for the current pandemic to pass. The latency, throughput, and scale are important factors for any use case deployment, but battling out current challenges pertaining to procurement, proving the ROI for a use case, cultural alignment with DevOps teams, adoption of microservices, and seamless use of orchestration platforms remain crucial for 5G's success in India. **EFY**